

B.E. INFORMATION TECHNOLOGY 2ND YEAR-2ND SEMESTER EXAMINATION- 2025**Subject: Graphics and Geometric Modeling**

Time: 3 hrs.

(Note: Answers of all parts of a question should be written together.)

Full Marks: 100

CO1 (15)	Q.1 Answer (a) and any one from (b) and (c): a. Consider a circle with center at (20, 15) and radius 25 units. Using both the Bresenham and Midpoint algorithms, generate all points of the circle in the 1st octant and calculate the time complexity of each algorithm based on the number of pixel calculations needed. 7+1 b. i. Consider a non-interlaced raster monitor with a resolution of n by m (m scan lines and n pixels per scan line), a refresh rate of r frames per second, a horizontal retrace time of $t_{horizontal}$ and a vertical retrace time of $t_{vertical}$. What is the fraction of the total refresh time per frame spent in retrace of the electron beam? ii. How long would it take to load a 640 by 480 frame buffer with 12 bits per pixel, if 10^5 bits can be transferred per second? How long would it take to load a 24-bit per pixel frame buffer with a resolution of 1280 by 1024 using this same transfer rate? 3+4 c. i. Use the midpoint method to derive decision parameters for generating points along a straight-line path with the slope in the range $0 < m < 1$. Show that the midpoint decision parameters are the same as those in the Bresenham line algorithm. ii. Suppose you have a system with a 12 inch by 14 inch video monitor that can display 120 pixels per inch. If memory is organized in one byte words, the starting frame buffer address is 0, and each pixel is assigned 4 bits of storage, what is the frame buffer address of the pixel with screen coordinates (x, y)? 3+4
CO2 (25)	Q.2 Answer (a) and any one from (b) and (c): a. i. Design a cubic Bezier curve represented by the control points $p_1(-8, 0, 0)$, $p_2(-4, 5, 0)$, $p_3(4, -5, 0)$, and $p_4(8, 0, 0)$. ii. A cubic splined curve has start and end point at (20, 0) and (7, 2) respectively. The tangent vectors at start and end points meet at (18, 9). Determine the Hermit cubic spline curve and hence find out its mid-point. iii. Compute and sketch the Bézier blending functions for a cubic Bézier curve. Indicate their intersection points and discuss their properties. 6+7+6 b. i. A B-spline curve has the knot vector $\{0, 0, 0, 0, 1, 2, 3, 4, 4, 4, 4\}$. Determine the degree of the curve and the number of control points used to define it. ii. If a B-spline curve has order 4 and 6 control points, how many segments does it contain, and how many control points influence each segment, and what is the degree of the curve? 3+3 c. i. What do C^0, C^1, and C^2 continuity mean in spline interpolation, and how do they affect the smoothness of the curve? ii. Construct the full knot vector for a B-spline curve of degree 3 using 4 control points. 3+3
CO3 (25)	Q.4 Answer (a) and any one from (b) and (c): a. The lower left-hand corner and upper right-hand corner of a rectangular window are at (-3, 1) and (2, 6) respectively. Using Sutherland-Cohen algorithm, check and find the visibility portion of the line segments AB, BC and CD for A(-4, 2), B(-1, 7), C(-1, 5), and D(-2, 3) in the clipping window. 10 b. i. A triangle having vertices (0, 3, 3), (3, 3, 0), and (3, 0, 3), is rotated 60° around a point (2, 2, 2). Find the new coordinates of the rectangle. ii. Prove that the multiplication of transformation matrices for each of the following sequences is commutative: i) Two successive rotations, ii) Two successive translations, iii) Two successive scalings. iii. Describe vanishing points and their function in perspective systems: 6+6+3 c. i. Vertices of a triangle are (2, 4, 4), (1, 4, 2), and (4, 4, 2). Find the new coordinates of the triangle when it is rotated 45° about X, Y and Z-axis (separately). ii. Prove that the product of two 2D rotation matrices $R(\theta_1)$ and $R(\theta_2)$ results in a rotation matrix $R(\theta_1 + \theta_2)$. Explain multi-point perspective and its typical applications. iii. Reflection is a special case of rotation- justify your answer. How does translation affect the rotation of an object about a point? 6+(3+2)+(2+2)

CO4 (15)	Q.4 Answer any one: a. - Find out RGB and YIQ color vectors whose equivalent CMY vector is (0.2, 0.4, 0.6). What does this say about color inversion? Among HSV and HLS, which is better for image editing and why? - Discuss the strengths and weaknesses of rasterization and ray tracing techniques. Compare the use of these techniques in real-time rendering and high-quality rendering. - How do half-toning and dithering technique differ in terms of their visual effects and application? - Discuss the differences you might expect to see in the appearance of specular reflections modeled with $(N.H)^n$ compared to specular reflections modeled with $(V.R)^n$. (3+3)+(2+3)+2+2 b. - Explain how to convert a color from the RGB model to both CMY and YIQ models. What are the essential components of the 3D rendering? - Compare ambient light, diffuse reflection, and specular reflection in illumination models in terms of description, appearance, depending factors, strength, and weakness. - Differentiate between Ray-casting, Ray tracing and Rediosity as rendering approach, strength, and weakness. - Describe the role of depth peeling in managing transparency in 3D rendering. How does it improve the rendering of complex transparent objects? (2+2)+4+3+(2+2)
CO5 (13)	Q.5 Answer any one: a. - Compare the structures and applications of quadtrees and octrees in spatial data representation. State the role of a scene graph in organizing and managing objects within a 3D modeling environment. - Explain the concept of content modeling in UX/UI design. In what ways can wireframes support or enhance the content modeling process? Provide relevant examples to justify your explanation. - Compare the usefulness of polygon meshes with CSG in representing complex solids. What are the limitations of using wireframe models in 3D modeling and visualization? Discuss how CSG operations can be applied in 3D modeling with the help of appropriate examples. 4+4+5 b. - Find the total length after n^{th} order of Koch Curve fractal where number of segments and scaling factor are 6 and $1/3$ respectively. Also find the fractal dimension. - Describe the purpose of the firing plane in ray-casting when used to implement CSG operations. How is ray-casting utilized in the implementation of CSG operations? Illustrate its role with a typical use case. - How to raycast octree for closest leaf node? What is the fastest way to insert and move a rectangle object into a linear octree? 4+5+4
CO6 (7)	Q.6 Answer any one: a. In what ways does VR enhance user experience compared to traditional 3D environments? How can VR be integrated with AR to create a mixed reality experience? Why might AR be a more suitable option for creative professionals compared to VR? Support your argument with examples. 2+2+3 b. Outline the standard process involved in creating a morphing sequence. What functions does a 3D visualizer perform in multimedia applications? List and briefly describe common authoring metaphors used in multimedia systems. 2+2+3

Course Outcomes

- CO1: Describe fundamental terms of computer graphics and input-output devices of Computer Graphics and Explain the algorithms for Raster scan Graphics.
 CO2: Illustrate curve and surface representation and drawing algorithms.
 CO3: Compute 2D and 3D Geometric transformations and apply viewing algorithms in typical cases.
 CO4: Describe color representation and rendering mechanisms of images.
 CO5: Describe Object representation and Illustrate Geometric Modeling.
 CO6: Discuss various topics of advanced computer graphics such as fractals, animations, morphing etc.

BACHELOR OF INFORMATION TECHNOLOGY ENGG. EXAMINATION, 2025

2nd year, 2nd semester

Software Engineering

Full Marks: 100

Time: 3 Hours

Read the questions carefully and answer them.

CO1: Answer Q1 and any one from the rest.

Full marks: 20

1. (a) You have to develop a software product for a very large steel company with multiple departments. They have a complex legacy system. Data migration from this legacy system is not an easy task and would take a considerable time. What process models will you choose? Justify your answer.
- (b) With a clear diagram explain that model of software development. (4+6)

2. (a) Mention any five principles of software engineering.
- (b) Which phase consumes maximum effort for developing a typical software product? Explain the reason.
- (c) Among all phases of software development, an undetected error from the design phase that ultimately gets detected during the system acceptance test, costs the maximum. Explain the reason. (4+3+3)

3. (a) With a clear diagram explain Incremental Development Model for software development. When we select Incremental model?
- (b) Differentiate between the waterfall model and the incremental model. (6+1+3)

CO2: Answer any two questions

Full marks: 20

4. (a) Why is the SRS document known as the black-box specification of a system? What do you understand by traceability of requirements?
- (b) What is a Feasibility Study? Why it is so important?
- (c) Explain about the various types of system modeling. (4+3+3)
5. (a) What are the relative disadvantages of using either the LOC or the function point metric to measure the size of the software?
- (b) If an organic project requires 160 person-months (PM) as its effort to complete the project, calculate the size and the development time for basic COCOMO. (5+5)
6. (a) Consider the following Temperature Monitoring System. This system senses temperature continuously and displays it in a LCD. It also takes input from a switch which tells it whether to display in the temperature in C or F. It also activates an alarm when the temp goes out of range.
Find out the activities for the above project, assign suitable time in each activity and draw an Activity Network for the same. Find out the critical path?
- (b) Draw the Gantt chart representation of the project. ((4+2)+4)

[Turn over

CO3: Answer any one questions**Full marks: 20**

- 7/ (a) What are the various types of Cohesion are there? Explain with examples.
 (b) "We cannot decompose a large problem into any smaller module infinitely." Explain your answer.
 (c) Draw a Use Case Diagram for the following system.
 A Library lends books and magazines to member, who is registered in the system. It also maintains the purchase of new books and magazines for the Library. A member can reserve a book or magazine that is not currently available in the library, so that when it is returned or purchased by the library, that person is notified. The library can easily create, replace and delete information about the books, members, and reservation in the system. The books transactions are stored in the database. The fine list while the member returns the book after the due date must be generated.

(6+4+10)

8. (a) Differentiate between structured analysis and structured design.
 (b) With one example explain the balancing a DFD.
 (c) What are the main shortcomings of Data Flow Diagram (DFD)?
 (d) Draw a Data Flow Diagram/ Class Diagram for the Library Management System described in question 7(c).

(3+4+3+10)**CO4: Answer any two questions****Full marks: 30**

- 9/ (a) Discuss with the help of appropriate examples:
 (i) Branch coverage criterion and (ii) Path coverage criterion
 (b) Which one of the following is the strongest structural test technique: Statement coverage-based testing, branch coverage-based testing, or condition coverage-based testing?
 (c) Discuss the difference between code inspection and code walk-through.

((4+4)+3+4)

- 10/ (a) Write a pseudo code for testing a grade calculation system that determines all scores from 90 to 100 as grade of A, 50 to 89 as P and less than 50 Fail. Find out the test cases for Equivalence Class Partitioning.
 (b) Find out the test cases for the boundary value analysis for the above system.
 (c) Draw the control flow graph for the above system and calculate "Cyclomatic Complexity".
 (d) What is symbolic debugger?

(3+3+(4+3)+2)

11. (a) What do you understand by the term system testing? How it is different from Integration testing.
 (b) How regression and stress tests are performed.
 (c) Differentiate between the Alpha and Beta testing.
 (d) Differentiate between verification and validation.

(4+4+4+3)**CO5: Answer the following question****Full marks: 10**

12. Write short notes on any two:
 (a) Software Quality Management.
 (b) People Capability Maturity Model.
 (c) Process based Quality Improvement Program.

(5+5)

Bachelor of Information Technology 2nd Year 2nd Semester Examination 2025
Sub: Computer Networks

EX/IT/PC/B/T/225/2025

Full Marks: 100

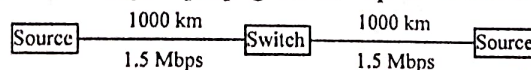
Time: 3 hours

Answer all questions. Answers must be brief and to the points.

1. (4 + 4 + 4 + 2 + 2)
- Describe what procedure might be involved inside the network in making a telephone connection. Now consider a personal communication service that provides a user with a personal telephone number. When the number is dialed the network establishes a connection to wherever the user is located at the given time. What functions must the network now perform to implement this service?
 - Standard containers are developed for the transportation of goods. These standard containers could fit on a train car, on a truck or in specially designed container ship. The standard size of the containers makes it possible to load and unload them much more quickly than non-standard containers of different sizes. Draw an analogy to packet switching communication networks. In your answer identify what might constitute a container and speculate on the advantages that may come from standard size information container.
 - Two musicians located in different cities want to have a jam session over a communication network. Find the maximum possible distance between the musicians if they are to interact in real time, in the sense of experiencing the same delay having each other as if they were 10 meters apart. The speed of sound is approximately 330 meters/second and assume that the network transmits the sound at the speed of light in cable 2.3×10^8 meters/second.
 - How does the network layer in a connection-oriented packet switching network differs from the network layer in a connectionless packet switching network?
 - Suppose the algorithms used to implement the operations at layer k is changed. How does this impact operations at layer $k - 1$ and $k + 1$?
2. (5 + 4 + 7)
- ATM uses an 8-bit CRC on the information contained in the header. The header has six fields: GFC (4bits), VPI (8bits), VCI (16bits), Type (3bits), CLP (1bit), CRC (8 bits). The generator polynomial is $x^8 + x^2 + x + 1$. Find the CRC bits if the GFC, VPI, Type, and CLP fields are all zero and the VCI field is: 00000000 00001111. Assume the GFC bits are at the MSB position.
 - Two computers are connected by an intercontinental link with a one-way propagation delay of 100 ms. The computers exchange 1 MB files that they need delivered in 250 ms or less. The transmission lines have a speed of R Mbps, and the bit error rate is 10^{-8} . Design a transmission system by selecting the bit rate R , the ARQ protocol, and the frame size.
 - A bidirectional link uses selective repeat ARQ with a window size 4. Assume all frames are one unit long, use a time-out value of 2, the one-way propagation delay is 0.5 time unit, the processing times are negligible, and the ACK timer is one unit long. Assuming station A and B begin with their sequence numbers set to zero, show the pattern of transmissions and associated state transitions for the following sequences of events: a) Station A sends six frames in a row, starting at $t=0$. All frames are received correctly. b) Station A sends six frame in a row, starting at $t=0$. All frames are received correctly, except frame 3 is lost. c) Station A sends six frames in a row, starting at $t=0$. Station B sends six frames in a row starting at $t=0.25$. All frames are received correctly.

OR

A 64 KB message is to be transmitted from the source to the destination as shown below. The network limits packets to a maximum size of 2 KB and each packet has a 32B header. The transmission lines in the network have a bit error rate of 10^{-6} , and Stop-and-Wait ARQ is used in each transmission line. How long does it take on the average to get the message from the source to the destination? Assume that the signal propagates at a speed of 2×10^5 km/second.



[Turn over

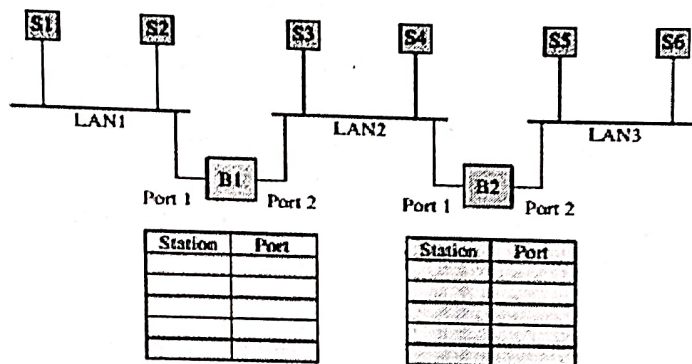
[2]

Ex IT/PC/B/T/225/2025

3.

(3 + 3 + 4 + 6)

- i. What is the length of a contention slot in CSMA/CD for a 2-km twin-lead cable (signal propagation speed is $0.82 \times 2.46 \times 10^8$ m/s) and a 40-km multimode fiber optic cable (signal propagation speed is $0.65 \times 1.95 \times 10^8$ m/s)?
- ii. Provide a brief answer and explanation for each of the following questions: (a) under a light load, which LAN has a smaller delay: Ethernet or Token Ring? (b) Under a high load, which LAN has a smaller delay: Ethernet or Token Ring?
- iii. Six stations (S1-S6) are connected to an extended LAN through transparent bridges (B1 and B2), as shown in the figure below. Initially, the forwarding tables are empty. Suppose the following stations transmit frames: S2 transmits to S1. S5 transmits to S4. S3 transmits to S5. S1 transmits to S2, and S6 transmits to S5. Fill in the forwarding tables with appropriate entries after the frames have been completely transmitted.



- iv. M terminals are attached by a dedicated pair of lines to a hub in a star topology. The distance from each terminal to the hub is d meters, the speed of the transmission lines is R bits/sec, all packets are of length 12,500 bytes, and the signal propagates on the line at a speed of 2.5×10^8 m/s. For the four combinations of the following parameters ($d = 25$ meters or $d = 2500$ meters; $R = 10$ Mbps or $R = 10$ Gbps), compare the maximum network throughput achievable when the hub is implementing slotted ALOHA and CSMA/CD.

OR

A 1-km-long, 10-Mbps CSMA/CD LAN (not 802.3) has a propagation speed of 200 m/μsec. Repeaters are not allowed in this system. Data frames are 256 bits long, including 32 bits of header, checksum, and other overhead. The first bit slot after a successful transmission is reserved for the receiver to capture the channel in order to send a 32-bit acknowledgement frame. What is the effective data rate, excluding overhead, assuming that there are no collisions?

4.

(4 + 4 + 2 + 3 + 3)

- i. Suppose that an interactive video game is accessed over a communication network. What requirements are imposed on the network if the network is connection-oriented? Connectionless?
- ii. Explain how the use of hierarchy enhances scalability in distance vector routing and link state routing algorithm.
- iii. Describe two major differences between the ECN method and the RED method of congestion control.
- iv. How the following policies at the data link layer affects congestion in the network?

Retransmission, Out-of-Order packet caching.

- v. Suppose a network uses link state routing, Explain what happens if:
- The router fails to claim a link that is attached to it.
 - The router claims to have a link that does not exist.

5. (4 + 4 + 2 + 4 + 6)

- Suppose that network addresses are scarce and are assigned so that they are not globally unique, in particular, suppose that the same block of addresses may be assigned to multiple organizations. How can the organizations use these addresses? Can users from two such organizations communicate with each other?
- What are the pros and cons of en-route reassembly versus reassembly at the final destination?
- Can we have a single bit subnet? Explain.
- A router is blasting out IP packets whose total length is 1024 bytes. Assuming that packets live for 10 sec, what is the maximum line speed the router can operate at without danger of cycling through the IP datagram ID number space?
- Suppose the population of the world is 6 billion people and that there is an average of 1000 communicating devices per person. How many bits are required to assign a unique host address to each communicating device? Suppose that each device attaches to a single network and that each network on average has 10000 devices, how many bits are required to provide unique network ids to each network. What is the size of the address?

OR

- ✓ An ISP is granted a block of addresses starting with 190.100.0.0/16. The ISP needs to distribute these addresses to three groups as follows: The first group has requested 16300 addresses. The second group has requested 16400 addresses. The third group has requested 8192 addresses. Show what addresses are assigned to which group? How many addresses are left to the ISP?

6. (3 + 3 + 4 + 6)

- ✓ Imagine that a two-way handshake rather than a three-way handshake were used to set up connections. In other words, the third message was not required. Are deadlocks now possible? Give an example or show that none exist.
- ✓ Lost TCP acknowledgements do not necessarily force retransmissions. Explain why?
- Suppose that the TCP congestion window is set to 18 KB and a timeout occurs. How big will the window be if the next four transmission bursts are all successful? Assume that the maximum segment size is 1 KB.
- Suppose that a TCP source (with unlimited amount of information to transmit) begins transmitting onto a link that has 1 Mbps in available bandwidth. Sketch congestion window versus time trajectory. Now suppose that another TCP source (also with unlimited amount of information to transmit) begins transmitting over the same link. Sketch the congestion window versus the time for the initial source.

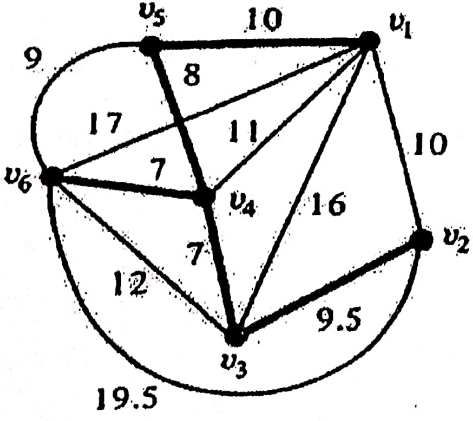
OR

What problem does TCP Reno have with multiple packet losses from a window? Give an example to illustrate.

Bachelor Of Engineering In Information Technology
2nd Year 2nd Semester, Semester Examination, 2025
Subject Name – Graph Theory & Combinatorics (IT/PC/B/T/224)

Time: 3Hrs.

Full Marks=100

CO1 [10]	Q1. (i) Consider a graph with four vertices labeled A, B, C, and D. The graph contains the following undirected edges: A is connected to B, A is connected to C, B is connected to C, and C is connected to D. Based on this graph, draw the corresponding structure. Find its adjacency matrix and incidence matrix. Additionally, compute the path matrix of length 2. (ii) Prove that "If a graph has exactly two vertices of odd degrees, there must be a path joining these two vertices". (iii) Find the maximum and minimum number of edges in a 3-component graph having 10 vertices. OR In a group of twelve (12) persons, is it possible that each person has a friendship with exactly five (5) others in the group? Justify this. [3+4+3=10]
CO2 [20]	Q2. (i) Prove that every tree has either one or two centers. (ii) Prove that any subgraph g of a connected graph G is contained in some spanning tree of G if and only if g contains no circuit. (iii) In an electric network, there are 5000 cables connecting 300 nodes. The network is currently fully connected, and each cable represents an edge in the graph. Calculate the minimum number of cables that must be removed to form a spanning tree in the network, ensuring the network remains connected but without any cycles. (iv) Using Prim's Algorithm, find the minimum spanning tree (MST) of the given graph (Figure: 1). Show all the steps involved and calculate the total weight of the MST.  Figure: 1 [4+5+5+6=20]
CO3 [20]	Q3. (i) Consider the following two undirected graphs: G_1: Vertices = {A, B, C, F}, Edges = {AB, AC, BC, CF} G_2: Vertices = {C, D, E, G}, Edges = {CD, DE, EG, GC}

- (a) Perform the union operation on G_1 and G_2 .
- (b) From the union graph, delete vertex D.
- (c) Identify at least one cut vertex and one cut edge in the union graph, and justify how their removal affects the structure and connectivity of the graph.
- (d) On the union graph, perform a fusion operation by merging vertices A and D, and update the graph accordingly.
- (ii) "The vertex connectivity of any graph G can never exceed the edge connectivity of G". Justify this.
- [(3+3+4+4)+6=20]

CO4
[20]

Q4.

(i) "A graph is 2-colorable if it is bipartite and every cycle has an even length." Justify this with a suitable example.

(ii) Consider a graph $G=(V, E)$ where $V=\{A,B,C,D,E,F\}$ and $E=\{(A,B),(A,C),(A,D),(B,C),(C,D),(D,E),(E,F),(C,F)\}$. Find the **chromatic number** of the graph. (Mention all steps properly)

(iii) Consider the graph $G=(V,E)$ where $V=\{A,B,C,D,E,F,G,H\}$ and $E=\{(A,B),(B,C),(C,D),(E,F),(F,G),(G,H)\}$

(a) Draw the graph G.

(b) Determine the **maximum independent set** of G.

(c) What is the size of this maximum independent set? Justify your answer with a proper explanation.

(iv) Given the graph G with the following details:

- Vertices = $\{A, B, C, D, E\}$
- Edges = $\{AB, AC, AD, BC, BD, CD, DE, CE\}$

Determine whether the graph is planar or non-planar. Justify your answer using **Kuratowski's Theorem**.

(v) Find out how many vertices can be matched using maximum matching in the bipartite graph algorithm of the following graph given below (Figure: 2).

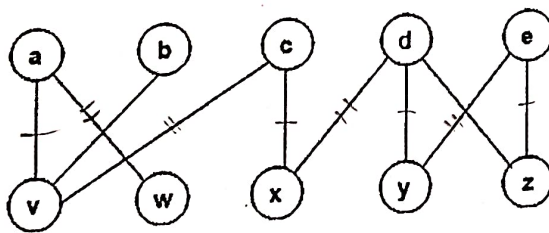


Figure: 2

[3+4+(2+3+2)+3+3=20]

CO5
[20]

Q5.

(i) How many license plates can be made using either three uppercase English letters followed by three digits or four uppercase English letters followed by two digits?

(ii) How many cards must be selected from a standard deck of 52 cards to guarantee that at least three cards of the same suit are chosen?

	(iii) A club has 25 members. How many ways are there to choose four members of the club to serve on an executive committee? (iv) There are six different candidates for governor of a state. In how many different orders can the names of the candidates be printed on a ballot? [4+5+6+5=20]
CO6 [10]	Q6. (i) Find the generating functions of the following sequences in closed form. $\langle 1, a, a^2, a^3, \dots \rangle$ (ii) Find out the solution of the recurrence relation given below with $a_0 = 2$ and $a_1 = 7$. $a_n = a_{n-1} + 2a_{n-2}$ [5+5=10]

CO1: Explain and discuss the concept of different types of Graphs with fundamental properties and express different types of matrix representation. (K2)

CO2: Illustrate different types of trees such as (i) rooted tree (ii) spanning tree etc, and explain their properties. (K3)

CO3: Apply operations like Union, Deletion, and decomposition of graphs and illustrate Cut vertex and Cut edge and their properties. (K3)

CO4: Illustrate planar graph and their properties and Graph Coloring and Matching. (K3)

CO5: Apply and evaluate basic counting rules, pigeon-hole principle and principle of inclusion-exclusion. (K3)

CO6: Apply and Solve problems using Generating Function and Recurrence Relations. (K3)

B.E. INFORMATION TECHNOLOGY SECOND YEAR SECOND SEMESTER - 2025

MATHEMATICS FOR IT-II

Time: 3 Hrs.

(Please use separate answer scripts for each part.)

Full Marks: 100

Part I

Marks : 60

CO1: Define and illustrate the concept of probability and random variables.

1. Answer any *three* questions.

[3 × 5 = 15]

- In a bolt factory machine M_1 , M_2 and M_3 manufacture respectively 25%, 35% and 40% of the total of their output with 5%, 4%, and 2% respectively are defective bolts. One bolt is chosen at random from the product and it is found defective. What is probability that it was manufactured in machine M_1 , M_2 , and M_3 ?
- A random variable X has the pdf: $f(x) = cx^2$ for $0 \leq x \leq 1$ and 0 otherwise. Find the value of c , and $P(0 \leq X \leq 1/2)$, $P(1/4 < X < 3/4)$. Also find the cdf of the given pdf.
- i) For what value of "a" will the function $f(x) = ax$; $x = 1, 2, 3, \dots, n$ be the pmf of a discrete random variable X . Find the variance of X .
ii) Obtain the expectation of the number of tails preceding the first head in an indefinite series of the tosses of the same coin.
- An unbiased coin is thrown three times. If the random variable X denotes the number of heads obtained, find the cumulative distribution function of X . Also, find the mode of X and the probability that at least two heads are obtained.

CO2: Discuss the concept of the continuous and discrete distribution along with its mean and variance and illustrate with different distributions.

2. Answer any *four* questions (where, F or ϕ carries their usual meaning).

[4 × 5 = 20]

- Derive the expressions of mean and variance of Binomial distribution with given parameters.
- i) A die is thrown 1200 times. Find the probability that the number of sixes lies between 190 and 200. (Given that the area of standard normal curve between 0 and 0.78 is 0.2823 and between 0 and 0.81 is 0.2910.)
ii) A discrete random variable X follows Poisson distribution such that $P(X = 0) = P(X = 1)$. Find the variance of the distribution and $P(X > 0)$.
- i) In a sample of 120 workers in a factory, the mean and s.d. of wages were INR 11.35 and INR 3.03 respectively. Find the number of workers getting wages between INR 9 – INR 17 in the whole factory, assuming that the wages are normally distributed. (Given: area under standard normal curve from 0 - 0.78 is 0.2823 and 0 - 1.86 is 0.4686).
ii) Show that the probability that the number of heads in 400 tosses of a fair coin lies between 180 to 220 is approximately $2\phi(2) - 1$.
- i) 2% of the items made by a machine are defective. Find the probability that 3 or more items are defective in a sample of 100 items (Given $e^{-3} = 0.0498$).
ii) If X and Y are independent Poisson variates, find the conditional distribution of X , given $X + Y$.
- i) In an examination, marks obtained by the student in C, JAVA and Python are independently and normally distributed with mean 50, 52, 48 and s.d.s. 15, 12, 16 respectively. Find the probability that total marks of a student are between 90 and 180 (Given $\phi(1.2) = 0.8849$ and $\phi(2.4) = 0.9918$).
ii) A variable X follows Poisson distribution with mean 16. Find $P(X \geq 20)$. Given $F(0.875) = 0.8092$.

CO3: Explain the concept of random sampling and measure central tendency, correlation, and regression.

3. Answer any *three* from {a, b, c, d} and *one* from {e, f}.

[3 × 4 + 3 = 15]

- Show that the mean and standard error of sample mean ($\bar{x}$) from simple samples of size n are $E(\bar{x}) = \mu$ and $S.E.(\bar{x}) = \sigma/\sqrt{n}$.
- Two lines of regression are given by $x + 2y = 5$ and $2x + 3y = 8$ and $\sigma_x^2 = 12$. Calculate mean of x , σ_y , and r .
- If m_r and α_r denote the r^{th} central moments of x and of the standard variable $z = (x - \bar{x})/\sigma$ respectively, where σ is the s.d. of x , then show that $\alpha_r = m_r/\sigma^r$ and justify "if all standardized moments α_r for $r > 2$ are 0, then x must follow a normal distribution".
- The regression equation calculated from a given set of observations are $x = -0.2y + 4.2$ and $y = -0.8x + 8.4$. Calculate r and estimated value of y when $x = 4$.

[Turn over

- e. If m_2' and m_2 are respectively the second moment about an arbitrary origin "a" and that about mean of x, then show that $m_2' = m_2 + d^2$, where $d = \bar{x} - a$.
- f. The mean of a certain normal distribution is equal to the standard error of a sample of 25 from the distribution. Find the probability that the mean of a sample of 49 from the distribution will be negative. (Given that area under standard normal curve to the left of the ordinate at 1.4 is 0.9192.)

CO4: Explain and illustrate Estimation of Parameters and curve fitting.

4. Answer *all* questions.

[5 + 5 = 10]

- a. The profits y (Rs. in lakh) of a certain company in the x^{th} year of its life are given y: (2.18, 2.44, 2.78, 3.25, 3.83) for x: (1, 2, 3, 4, 5). Obtain the profit at 6th year by fitting a second degree parabola to the given data.
- b. Fit an exponential model of the form $v = ab^u$ to the following data. u: (2, 3, 4, 5, 6, 7) with v: (640, 512, 410, 328, 262, 210) and find the value v at $u = 0.5$ and $u = 8.5$.

B.E. INFORMATION TECHNOLOGY SECOND YEAR SECOND SEMESTER - 2025

MATHEMATICS FOR IT-II

Use Separate Answer scripts for each Part

Part II

Marks : 40

CO5: Understand the basics of game theory and Inventory Control and some simple applications

5. Attempt (a or b), c and d

[6 + 4 + 10 = 20]

a. Which competitive situation is called a game? What are assumptions made in the theory of games? What is the maximin criterion of game?

OR

b. Find solution of the following game

	Player II			
Player I	6	8	7	5
	8	9	5	7
	7	4	6	3
	6	6	8	7
	8	9	8	7

c. Let $[a_{ij}]_{m \times n}$ be the payoff matrix and $(p_1, p_2, \dots, p_m)$ and $(q_1, q_2, \dots, q_n)$ be the probabilities with which players A and B select their strategies $(A_1, A_2, \dots, A_m)$ and $(B_1, B_2, \dots, B_n)$, respectively. If V is the value of the game, then express the loss of the player B.

d. Consider the inventory model as defined below. Explain the model with a diagram and derive economic order quantity, optimal stock level and total variable inventory cost, optimum shortage level and total cycle time.

- | | |
|---|---|
| i. One type of item in the inventory | v. Demand is known, constant |
| ii. Replenished instantaneously and single delivery | vi. Lead time is constant and known |
| iii. Shortage is allowed | vii. Purchase cost and reorder cost same, no discount |
| iv. Carrying cost per year is same | |

CO6: Explain and illustrate the M/M/1 queue and Project Management

6. Attempt any three (3) from {a, b, c, d, e} and one from {f, g}

[3 × 4 + 8 = 20]

a. For an activity (i, j) define total float time and free float time. State and prove the relation between total float time and free float time of an activity.

b. For a critical activity (i, j), prove that $E_j = L_j$ and $E_i = L_i$ are necessary condition but not sufficient.

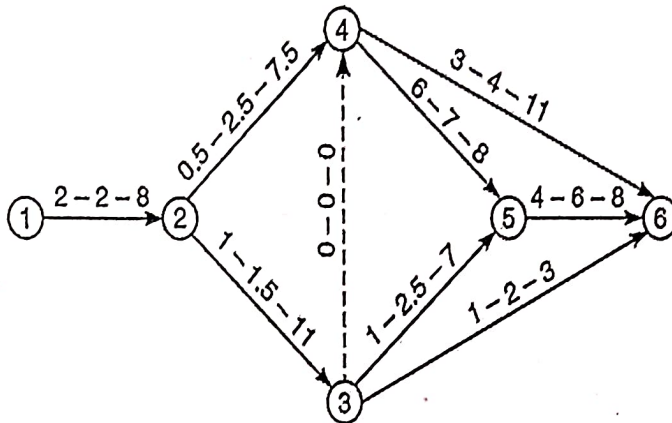
c. For a project, if you have two critical paths, as a project manager which path do you follow, when the problem is solved by i) CPM; ii) PERT method. Justify your answer.

d. What is an isolated event? Which distribution is used in the analysis of the isolated event? Write the expression of the distribution with explaining the meaning of the symbols.

e. A queuing model is represented by (a/b/c) : (d/e/f). Define the symbols a – f.

[Turn over

- f. Consider the network shown below. The three time estimates for the activities are given along the arrows. Determine the critical path. What is the probability that the project will be completed in 21 days? [Z score table is given below]



Standard Normal Distribution: Table Values Represent area to the left of the Z score										
Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891

- g. In a queuing model, the steady state equation for this system is given below, where the symbols have usual meaning.

$$P_n'(t) = \lambda P_{n-1}(t) + \mu P_{n+1}(t) - (\lambda + \mu) P_n(t), \quad n \geq 1$$

$$P_0'(t) = \mu P_1(t) - \lambda P_0(t), \quad n = 0$$

Assuming above equation prove that $P_n = \left(1 - \frac{\lambda}{\mu}\right) \left(\frac{\lambda}{\mu}\right)^n$

BE in INFORMATION TECHNOLOGY

2ND YEAR, 2ND SEMESTER EXAMINATION, 2025

OBJECT ORIENTED SYSTEMS

Full marks: 100

Total Time: 3 hours

CO1 [40 MARKS]	1. a) Justify the validity of the following statements with supporting arguments. Provide code snippet/s where/ if necessary. i) There is only one entry point to a program, irrespective of the number of <i>main()</i> methods in a class. ii) Method overriding is also known as late binding. iii) A non-static member method can be invoked within the static block of a class. iv) Reference data types do not store the value directly in a variable. v) Every member method in a Base class can be overridden in its Child class. vi) An interface solves the diamond problem automatically. vii) The <i>ensureCapacity()</i> method doubles the capacity of a <i>StringBuffer</i> object. viii) Two strings can be compared by <code>==</code> operators. ix) A private member method in a <i>Base</i> class cannot be overridden in its <i>Child</i> class. x) Two different packages should never have classes having same name. b) Fill up the blanks with appropriate phrases. Hence justify the validity of each of the completed sentences with proper arguments. Provide code snippets where necessary. i) The difference between the keywords <i>final</i> and <i>finally</i> is _____. ii) Classes in a package must be declared as _____, otherwise they cannot be _____. iii) Static methods in a Base class can be overridden, but it results in method _____. iv) A _____ array is called a jagged array if _____. v) An abstract class may not have a/an _____, but it must be _____. vi) Run time polymorphism occurs only for _____ of a class, but not for _____. vii) The difference between checked and unchecked exception is _____. viii) The parameterized constructor of the Base class must be called from _____ by _____. ix) The <i>equalsIgnoreCase()</i> and <i>compareToIgnoreCase()</i> methods differ by _____. x) Two strings can be concatenated by _____ or _____. [(2x10)+(2x10)=40]
CO2 [25 MARKS]	2. a) Distinguish between each of the following pairs: Use supporting code snippets. i) <i>sleep()</i> and <i>wait()</i> ii) <i>interrupt()</i> and <i>interrupted()</i> iii) <i>readLine()</i> and <i>nextLine()</i> iv) <i>nextInt()</i> and <i>hasNextInt()</i> b) Fill up the blanks with appropriate phrases. Hence justify the validity of each of the completed sentences with proper arguments. Provide code snippets where necessary. i) It is better to create threads by _____ than _____. ii) Synchronized _____ ensure more concurrency than synchronized _____ because _____. iii) Race around condition occurs in _____ and is eliminated by _____. iv) Deadlock can be prevented by _____ so that no _____ occurs. v) Inter thread communication is achieved by _____.

[Turn over

	c) Write a Java class having a method that opens a text file <i>a.txt</i> in read mode, select all the palindrome words and write them into another file <i>b.txt</i>. Hence count the number of such words in the file. Or, Write a Java class having a method that opens a text file in read mode, select all the words containing atmost one vowel and write them into another file. Hence count the number of such words in the file. Here the file names should be taken as input from the user. [(2x4)+(2x5)+7=25]
CO3 [20 MARKS]	3. a) What special purpose is achieved by Java Reflection? b) Write suitable code snippets to invoke the <i>charAt()</i>, <i>substring(int,int)</i> and <i>indexOf()</i> methods of <i>String</i> class using reflection. Hence show the output with proper examples. c) Define a generic class having 3 overloaded versions of a generic method. One of them should be static. Now show with a suitable program how to invoke these methods from <i>main()</i>. d) Define a generic class with a member method <i>highest_occurence(A)</i> that take arrays A of different datatypes (int, float, char) as input and finds the highest occurring element. Example: An array contains 1,2,1,3,4,3,2,1,4,5,1,8,9,1: Highest occurring element is 1. Or, Invoke any 4 methods of <i>StringBuffer</i> class (which are not available for <i>String</i> class) using reflection. Hence show the output of each of them with suitable example. [2+6+5+7=20]
CO4 [10 MARKS]	4. a) Illustrate how the different parts of a class are represented in a class diagram. b) Distinguish between each of the following pairs: (Any 3) i) Aggregation and composition ii) Synchronous and asynchronous transmission in sequence diagram iii) Include and extend relations iv) Generalization and inheritance c) Discuss multiplicity and role related to class diagram of UML. [2+(2x3)+2=10]
CO5 [5 MARKS]	5. Discuss singleton design pattern. Or, Explain iterator design pattern. Or, Explain command design pattern.

Course Outcomes:

CO1: Differentiate different object oriented programming language and Solve problems by developing Java programs using (i) classes, (ii) inheritance, (iii) nested classes and (iv) exceptions.

CO2: Solve problems using thread programming and Input-Output.

CO3: Develop programs using advanced programming paradigms: (i) Introspection capabilities, (ii) Generic Programming.

CO4: Model and Sketch software systems by using different artifacts of Unified Modeling Language.

CO5: Explain and illustrate basics of design patterns by developing programs.

Question Booklet No.

401021

Question Booklet Code

A

Environmental Studies (EVS)
1ST YEAR 2ND SEMESTER EXAMINATION 2024
(COMPULSORY)

(Do not open this QUESTION BOOKLET until you are asked to do so)

Maximum Marks : 100

Duration

(Use Ball point Pen to fill up this cover page of the Question Booklet)

Name of the Candidate : Arish Ar Exam. Roll No

Date of Examination : 24.04.25

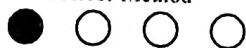
Signature of the Candidate : Arish Ar

IMPORTANT INSTRUCTIONS

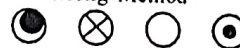
Candidates should read the following instructions carefully and fill all the required particulars before answering the questions:

- 1) The Question Booklet has paper seal pasted on it. Candidates should open the Question Booklet only when they are asked to do so by the Invigilator.
- 2) Immediately after breaking the seal the candidates must check that Question Booklet has 50 questions with multiple choice answers. **If there is any discrepancy, immediately report to the Invigilator for change of Booklet.**
- 3) Each Question Booklet has a Booklet Code A, B, C, D (printed at the Right Top Corner of the page) mentioned on it, which every candidate must carefully fill in the respective place provided for this in the OMR Answer Sheet. Failing to copy this series code in the OMR Answer Sheet will be liable to rejection of the OMR Answer Sheet.
- 4) Candidate must write his/her Name, Roll No., Date and then Sign on the front page of the Question Booklet at the appropriate places marked for this purpose.
- 5) Answers will have to be given on the Special OMR Answer Sheet supplied for this purpose. Question numbers progress from 1 to 50 with alternative answers shown as A, B, C and D.
- 6) Answer Sheets will be processed by electronic means. Hence, invalidation of Answer Sheet resulting due to folding or putting stray marks on it or any damage to the answer sheet as well as incomplete / incorrect filling of the answer sheet, will be the sole responsibility of the Candidate.
- 7) Please use **only blue or black ball point pen** to mark your answers.
- 8) While answering, choose only the Correct Alternative Answer from the four choices given alongwith the question and respond the same by shading the corresponding **OVAL** in the OMR Answer Sheet only. **Answer all questions (Each question carries 2 marks). There is no Negative mark.**
- 9) Darken completely only one **OVAL** which you think is correct as shown in the figure below:

Correct Method



Wrong Method



- 10) Mark the answers only in the space provided. Please do not make any stray marks on the answer sheet. Rough work must be done on the pages (Space for Rough Work) given at the end of the Question Booklet and/or inside the Question Booklet.
- 11) **Please hand over the OMR Answer Sheet to the Invigilator before leaving the Test/Examination Hall.**
- 12) Use of any electronic Device like Mobile, Calculator etc. is strictly prohibited, if found in possession, your candidature will be cancelled.



BACHELOR OF ENGINEERING EXAMINATION, 2024

(1st Year, 2nd Semester, Compulsory)

ENVIRONMENTAL STUDIES (EVS)

Time : Two hours

Full Marks : 100

Answer *fifty* questions (Each question carries 2 marks)

1. The polluted water is one which
 - a) contains pathogenic bacteria
 - ☒ b) consists of undesirable substances rendering it unfit for drinking and domestic use
 - c) is safe and suitable for drinking and domestic use
 - d) is contaminated
2. Which of the following is not a water borne disease?
 - a) dysentery
 - b) cholera
 - c) typhoid
 - ☒ d) malaria
3. The most common cause of acidity in water is
 - ☒ a) carbon dioxide
 - b) oxygen
 - c) hydrogen
 - d) nitrogen
4. Disinfection of water results in
 - a) removal of turbidity
 - b) removal of hardness
 - ☒ c) killing of disease bacteria
 - d) complete sterilisation
5. Global warming will cause the following problem
 - a) Melting of glaciers
 - b) Raising of ocean level
 - c) Sinking of islands
 - ☒ d) All the above
6. The gas which destroys chlorophyll in plant leaves is
 - a) SO₃
 - b) CO₂
 - ☒ c) SO₂
 - d) H₂S.
7. Which of the following three R's regarded as environment friendly
 - a) Reduce, Rebuild, Restrict
 - ☒ b) Reduce, Reuse, Recycle
 - c) Read, Register, Recall
 - d) Random, Reduce, Recall
8. Hydropower plants converts the kinetic energy of flowing water into _____
 - a) Mechanical energy
 - b) Tidal energy
 - c) Wind energy
 - ☒ d) Electricity

9. The suitable method for disinfection of swimming pool water is
- ultra violet rays treatment
 - lime treatment
 - by using potassium permanganate
 - chlorination
10. A renewable exhaustible natural resource is
- Forest
 - Coal
 - Petroleum
 - Minerals
11. Uses of forest includes
- Productive function
 - Protective function
 - Regulative function
 - All the above
12. Deforestation leads to reduction of ____
- Rain fall
 - Soil erosion
 - Ground water depletion
 - All the above
13. ____ fuels is the future fuel for mankind
- Biofuels
 - Hydrogen fuels
 - Biogas
 - Biomass energy
14. Process of burning of non-biodegradable solid waste is called ____
- Composting
 - Incinerations
 - Segregation
 - Sanitary land filling
15. Jim Corbett National Park was established in
- Uttarpradesh
 - Tamil Nadu
 - Gujarat
 - Uttarkhand
16. The main purpose of rainwater harvesting is recharge the
- Gardens
 - Wells
 - Cannels
 - Groundwater table
17. Disposal of any kind of electrical and electronic waste is called
- Waste
 - Domestic waste
 - Vegetable waste
 - e-waste
18. ____ Dam is situated in region of high seismicity
- Sardarsarovar dam
 - Nagarjuna sagar dam
 - Bhakra Nangal Dam
 - Tehri dam

[Turn over

19. Which natural factor cause drought

- a) Over cultivation
- b) Rain failure
- ☒ c) Over grazing
- d) None of the above

20. Which part of plant evaporates water

- ☒ a) Stomata
- b) Fruit
- c) Branch
- d) Root

21. In which year, U.N. Conference on human environment was held at Stockholm

- a) 1962
- ☒ b) 1972
- c) 1982
- d) 1992

22. The people should become aware of the dangers of _____

- a) Population explosion
- b) Deforestation and ozone layer depletion
- c) Industrialization and global warming
- ☒ d) All the above

23. IUCN Stands for

- ☒ a) International Union for Conservation of Nature
- b) International Union for Conservation of Nature and Natural Resources
- c) International Union Council for Nature
- d) International Union council for Conservation of Nature and Natural Resources

24. _____ is the second generation of pesticides which enter into the food chain

- a) Malathion
- b) Boric Acid
- c) Diazinon
- ☒ d) DDT

25. Which is called as autotrophs

- ☒ a) Green plants
- b) Animals
- c) Birds
- d) All the above

26. Endemic species are

- a) Rare species
- b) Cosmopolitan distribution
- c) Species localized in a specific region
- ☒ d) Critically endangered species

27. Which national park is famous for having Great Indian one Horned Rhino

- a) Rajaji National Park
- b) Jim Corbet National Park
- c) Bandipur National Park
- ☒ d) Kaziranga National Park

28. WWF stands for

- a) World Wide Fund
- ☒ b) World Wildlife Fund for Nature
- c) Wildlife Wide Fund for Nature and Natural Resources
- d) World Wild Fund

29. Ozone layer is present in _____

- a) Troposphere
- b) Ionosphere
- c) Lithosphere
- d) Stratosphere

30. Sulphur dioxide and nitrogen oxides are transported by prevailing winds they form secondary pollutants such as nitric acid vapour, droplets of sulfuric acid descends on earth surface is called _____

- a) Rain
- ☒ b) Acid rain
- c) Basic rain
- d) Ice rain

31. The Ozone layer in column is measured by _____

- a) Amstrong Unit
- b) Light year
- ☒ c) Dobson Unit
- d) Archimedis Unit

32. Activated carbon is used for _____

- ☒ a) disinfection
- b) removing hardness
- c) removing odours
- d) removing corrosiveness

33. Which of the following agency published the Red data book?

- ☒ a) IUCN
- b) NEERI
- c) NWAP
- d) CITES

34. Green House Gases are

- a) Carbon mono oxide, Carbon dioxide, Methane and Nitrogen
- b) Carbon mono oxide, Sulphur dioxide, Nitric oxide and Methane
- ☒ c) Carbon dioxide, nitrogen oxides, methane and CFC
- d) Carbon dioxide, Sulphur dioxide, CFC and methane

35. In the polar regions temperature rises is caused by _____ would have disastrous effects

- ☒ a) global warming
- b) Green house effect
- c) CFC
- d) Carbon monoxide

36. CPCB means

- a) Committee of Pollution Control Board
- ☒ b) Central Pollution Control Board
- c) Central Population Control Board
- d) Committee of Population Control Board

37. June 5th is observed as

- ☒ a) World Environmental Day
- b) World Forest Day
- c) World Wildlife Day
- d) World Population Day

[Turn over

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[Turn over

38. Which of the following is a primary pollutant
☒ a) Peroxy Acetyl Nitrate
 b) DDT
 c) Plastics
 d) SO_x
39. Air pollution can be controlled by
 a) Modification of industrial equipment
 b) Using eco-friendly Bio-diesel in the automobiles
 c) Fixing a green box at the end of the exhaust pipe of a car, which is used to create bio-fuels
☒ d) All the above
40. The important requirement of the aquatic life is ____
☒ a) Dissolved Oxygen
 b) Dissolved Chlorine
 c) Dissolved Nitrogen
 d) Dissolved Methane
41. The eggs of aquatic birds are not hatched due to presence of a _____ chemical in their food
 a) Malathion
 b) SO_2
☒ c) DDT
 d) None of the Above
42. Which of the following can be recycled many times
☒ a) Organic materials
 b) Aluminum
 c) Wood
 d) Plastics
43. How does organic material in the buried solid waste will decompose
☒ a) By the action of microorganisms
 b) By the soil particles
 c) By flow of water
 d) By the action of oxidation
44. Which of the following gases was responsible for Bhopal gas tragedy?
 a) Hydrogen fluoride
 b) Methyl chloride
 c) Hydrogen chloride
☒ d) Methyl Isocyanate
45. IPCC stands for
 a) Information Processing Command and Control
☒ b) Interprocess Communication and Control
 c) Intergovernmental Panel on Climate Change
 • d) Introgovernmental Panel on Climate Change
46. The most dangerous gas for ozone depletion are ____
☒ a) CFC
 b) CH_4
 c) N_2O
 d) All the above
47. The Water (Prevention and Control of Pollution) Act was enacted in the year
☒ a) 1974
 b) 1975
 c) 1986
 d) 1994

48. Stagnant water, which forms breeding sites of _____ mosquito is the most important factor in the spread of malaria

- a) Aedes
- b) Anopheles
- c) Culex
- d) Sabethes

49. Which oil can be used as a substitute for diesel?

- a) Castor oil
- b) Jatropha oil
- c) Cotton seed oil
- d) Flax seed oil

50. Father of green revolution in India

- a) M. S. Swaminathan
- b) Harikrishna Jain
- c) Vandana Shiva
- d) Dr. B. P. Pal